

SANTA CRUZ DE LA SIERRA, Bolivia

WMO: 852440

Lat: 17.645S

Lon: 63.135W

Elev: 1225

StdP: 14.06

Time Zone: -4.00 (W04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	49.7	52.0	31.9	27.5	59.5	35.8	32.3	61.2	31.1	76.0	29.1	75.6	12.7	170	0.450

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.5	95.1	74.3	93.4	74.7	91.7	75.0	79.0	87.8	78.3	87.0	77.8	86.4	17.1	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.0	147.4	82.5	75.9	141.5	81.2	75.5	139.7	80.8	43.6	88.3	42.9	87.5	42.3	86.6	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.1	25.5	22.7	DB	44.4	98.7	3.1	2.3	42.2	100.3	40.4	101.6	38.7	102.9	36.5	104.5	(4)
(5)			WB	40.7	81.4	3.2	1.3	38.4	82.3	36.5	83.0	34.7	83.7	32.4	84.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	75.7	79.7	78.9	78.3	75.8	71.1	69.0	69.2	73.0	76.6	78.7	79.1	79.3
	DBStd	6.68	3.04	3.14	3.42	5.13	6.36	7.05	7.35	6.91	6.70	5.13	4.30	3.42
	HDD50	1	0	0	0	0	0	0	1	0	0	0	0	0
	HDD65	143	0	0	0	3	20	44	48	19	7	1	1	0
	CDD50	9389	922	810	879	775	655	571	596	712	798	891	872	908
	CDD65	4054	457	390	414	328	211	164	177	265	356	426	423	443
	CDH74	33699	3835	2964	3108	2406	1290	944	1277	2412	3549	4107	4030	3777
CDH80	13392	1524	1081	1139	870	366	219	384	1034	1676	1866	1757	1476	
Wind	WSAvg	11.3	10.2	10.0	9.4	10.1	10.8	12.0	12.4	13.1	12.7	12.0	11.3	10.9
Precipitation	PrecAvg	55.1	7.2	7.3	5.3	4.3	3.7	3.0	2.2	1.3	3.0	4.0	5.4	7.3
	PrecMax	81.7	14.4	14.0	12.5	12.8	11.0	7.7	11.9	3.9	7.6	10.5	9.8	13.5
	PrecMin	37.0	3.6	3.1	3.0	1.1	0.4	0.0	0.1	0.0	0.0	0.8	1.3	2.3
	PrecStd	10.4	2.8	2.4	2.2	2.6	2.7	2.0	2.4	1.2	2.1	2.5	2.6	3.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	95.3	93.0	92.8	91.6	88.5	86.4	89.2	93.5	96.8	97.7	96.5	94.9
		MCWB	75.0	76.9	77.0	76.7	75.1	73.8	71.3	72.6	72.8	73.9	74.7	75.5
	2%	DB	92.8	91.1	90.0	89.7	86.3	84.5	86.2	91.3	94.8	94.9	93.4	91.7
		MCWB	76.0	76.7	76.3	76.3	74.5	73.0	71.5	71.7	73.3	74.6	74.9	76.3
	5%	DB	89.9	89.2	88.2	87.9	84.3	82.6	84.3	89.4	92.8	92.0	91.5	89.8
		MCWB	76.4	76.5	75.9	75.7	73.8	72.3	71.0	71.8	73.5	74.7	75.1	76.2
	10%	DB	87.9	86.4	86.3	85.8	81.9	80.4	82.1	86.4	89.8	89.8	89.4	87.8
		MCWB	76.0	76.1	75.7	75.4	72.5	71.3	70.2	71.0	72.7	74.5	74.9	76.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.9	79.9	79.1	78.7	77.6	76.4	75.0	76.1	77.6	78.5	79.0	79.5
		MCDB	88.9	88.1	87.4	87.0	84.6	82.7	84.1	87.5	89.8	89.2	89.3	88.7
	2%	WB	78.5	78.4	78.2	77.8	76.3	74.8	73.7	74.5	76.2	77.4	78.0	78.3
		MCDB	87.0	86.3	86.3	86.1	83.2	81.7	82.0	85.9	88.6	88.4	88.2	87.1
	5%	WB	77.8	77.7	77.5	77.0	75.2	73.7	72.6	73.5	75.1	76.6	77.2	77.6
		MCDB	85.9	85.2	85.3	85.2	81.7	79.7	80.4	84.6	87.5	87.4	87.1	86.1
	10%	WB	77.1	77.0	76.8	76.1	74.1	72.6	71.4	72.4	74.0	75.7	76.4	76.9
		MCDB	84.8	84.1	84.0	83.3	79.3	78.0	78.6	82.5	85.8	85.9	85.4	84.9
Mean Daily Temperature Range	5% DB	MDBR	15.5	14.8	15.4	16.5	15.3	15.6	18.2	20.8	20.5	18.5	17.7	15.6
		MCDBR	19.6	18.5	18.2	19.1	18.5	17.9	21.2	24.3	23.9	22.0	21.8	19.2
		MCWBR	7.1	6.9	6.9	7.7	8.1	8.4	9.0	9.1	8.3	7.4	7.9	6.9
	5% WB	MCDBR	16.2	15.6	15.9	16.7	15.6	15.5	18.2	20.9	20.8	18.6	17.6	16.1
		MCWBR	6.6	6.8	6.7	7.3	7.5	7.9	8.7	9.3	8.2	7.4	7.0	6.6
Clear-Sky Solar Irradiance	taub	0.402	0.397	0.387	0.375	0.372	0.370	0.386	0.504	0.631	0.522	0.427	0.410	
	taud	2.435	2.463	2.488	2.497	2.457	2.488	2.397	2.012	1.713	2.033	2.317	2.392	
	Ebn at Noon	299	298	293	285	274	269	267	243	224	259	290	297	
	Edh at Noon	39	38	36	33	32	30	34	53	76	57	44	41	
All-Sky Solar Radiation	RadAvg	1775	1652	1598	1442	1156	1063	1195	1405	1534	1665	1770	1724	
	RadStd	192	124	118	109	155	107	102	101	149	134	137	132	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.52	N/A	N/A	+1.04	N/A	N/A	N/A	N/A	+187	+169
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page